DAY 18 | POST-READ

Git, Pipelines and Docker: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
Git areas	Working folder, staging (git add), repository (git commit), remote (git push).
Branches	git switch -c feature/x. Keep main working; one branch per task.
Merge conflicts	Edit the marked lines, remove markers, git add, git commit.
Pull requests	Small, explained, CI green, reviewed kindly. Branch policies protect main.
Pipeline levels	Pipeline, stages, jobs, steps (scripts or tasks).
Stage control	dependsOn for order; condition to run only on main.
Docker terms	Dockerfile (recipe), image (package), container (running), registry (storage).
Dockerfile pattern	Slim base, WORKDIR, copy requirements, install, copy code, EXPOSE, CMD.
Docker commands	build -t, run -d -p, ps, logs, stop, rm.

Part B: Going beyond today

1. Good commit messages

```
Add priority filter to task list

Users asked to focus on urgent work. Adds ?priority= to GET /tasks
and a dropdown in the UI. Closes #42.
```

A short summary line, a blank line, then why. Your future self will thank you.

2. Layer caching

Docker caches each instruction. Copy requirements.txt and install before copying code, so code edits do not reinstall every package. Builds go from minutes to seconds.

3. Multi-stage Docker builds

```
FROM node:20 AS build
WORKDIR /web
COPY . .
RUN npm ci && npm run build

FROM nginx:alpine
COPY --from=build /web/dist /usr/share/nginx/html
```

Build with heavy tools, ship only the result. Ideal for React front ends.

4. Run as a non-root user

```
RUN useradd -m appuser
USER appuser
```

If an attacker breaks into the app, they do not get root inside the container.

5. Pipeline templates and variables

Large teams reuse steps with templates (template: steps/python-test.yml) and keep settings in variable groups, with secrets linked from Azure Key Vault.

Part C: Check yourself

- What is the difference between git add and git commit?
- How do you resolve a merge conflict?
- What is the difference between a stage and a job?
- Why copy requirements.txt before the application code in a Dockerfile?
- What does -p 8000:8000 mean in docker run?

Part D: Practice ideas for tonight

- Create a merge conflict on purpose in a practice repo and resolve it.
- Add a .dockerignore and compare image sizes before and after.
- Write a Dockerfile for your Day 10 Express API using node:20-slim.

Coming up on Day 19: containerising the full app, Azure App Service, Azure Container Apps and continuous deployment pipelines.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today